

PRANAV JAIN

University of Southern California

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[Website](#) , [Scholar](#)

RESEARCH INTERESTS

Geometry Processing, Physical Simulations, Computer Graphics, Discrete Differential Geometry, Numerical Analysis of Partial Differential Equations

EDUCATION

University of Southern California (USC) Doctor of Philosophy (PhD), Computer Science	August 2023 - Present <i>Los Angeles, USA</i>
New York University (NYU) Master of Science (MS), Scientific Computing	September 2021 - May 2023 <i>New York, USA</i>
Indraprastha Institute of Information & Technology Delhi (IIITD) Bachelor of Technology with Honors (B-Tech Hons), Computer Science and Applied Mathematics	August 2016 - August 2020 <i>New Delhi, India</i>

PUBLICATIONS

- Pranav Jain**, Ziyin Qu, Peter Yichen Chen, and Oded Stein. 2024. Neural Monte Carlo Fluid Simulation. In ACM SIGGRAPH 2024 Conference Papers (SIGGRAPH '24). [\[link\]](#)
- Liam Martin, **Pranav Jain**, Zachary Ferguson, Torkan Gholamalizadeh, Faezeh Moshfeghifar, Kenny Erleben, Daniele Panozzo, Steven Abramowitch, Teseo Schneider. A systematic comparison between FEBio and PolyFEM for biomechanical systems. In Computer Methods and Programs in Biomedicine, 2023. [\[link\]](#)
- Zachary Ferguson, **Pranav Jain**, Denis Zorin, Teseo Schneider, and Daniele Panozzo. High-Order Incremental Potential Contact for Elastodynamic Simulation on Curved Meshes. In ACM SIGGRAPH 2023 Conference Proceedings (SIGGRAPH '23). [\[link\]](#)
- Pranav Jain**, Munawar Hasan, Donghoon Chang. Spy based analysis of selfish mining attack on multi-stage blockchain. In Cryptology ePrint Archive, 2019. [\[link\]](#)

RESEARCH EXPERIENCE

Adobe Research <i>Research Intern</i> Advisor: Dr. Timothy Langlois, Dr. Danny Kaufman, Dr. Jui-hsien Wang — Researched on how to make generative-AI models more physics accurate	May 2025 - August 2025 <i>Seattle, USA</i>
University of Southern California <i>PhD Student</i> Advisor: Prof. Oded Stein — Researching on empirically proving convergence rates of Poisson problem using Physics Informed Neural Networks (PINN) with respect to the size of the architecture — Researched and published on a discretization-free fluid simulation method using combination of Monte-Carlo and neural fields	August 2023 - Present <i>Los Angeles, USA</i>

New York University*Research Assistant*

Advisor: Prof. Daniele Panozzo, Prof. Denis Zorin

- Researched and compared the simulation abilities of PolyFEM and FEBio in terms of accuracy, time and user inputs
- Introduced a high-order finite element formulation for elastodynamic simulation with contact handling

September 2021 - May 2023

*New York, USA***nTopology***Software Engineer Intern in Geometry Team*

Advisor: Suraj Musuvathy, Ranjeeth Mahankali

- Designed and implemented a method to efficiently stitch different CAD models to form one unified CAD structure

June 2022 - August 2022

*New York, USA***Freie Universität Berlin [\[link\]](#)***Research Intern*

Advisor: Prof. Konrad Polthier, Dr. Sunil Kumar Yadav

- Researched on a robust and parameter-tuning free method to denoise a point cloud

September 2020 - August 2021

*(Virtual) Berlin, Germany***Fields Undergraduate Summer Research Programme 2020 [\[link\]](#)***Research Intern*

Advisor: Prof. Thomas Uchida

- Researched on the problem of inverse mechanism synthesis – given a path, design a mechanism that traces the path

July 2020 - August 2020

*(Virtual) Toronto, Canada***Indraprastha Institute of Information & Technology Delhi [\[link\]](#)***Research Assistant*

Advisor: Prof. Kaushik Kalyanaraman

- Developed DECAGT, a software framework for discretizations of the objects and operators of exterior calculus

August 2018 - August 2020

*New Delhi, India***Indraprastha Institute of Information & Technology Delhi [\[link\]](#)***Undergraduate Thesis*

Advisor: Prof. Donghoon Chang

- Proved the selfish mining attack strategy on the Bitcoin blockchain for multiple mining pools
- Designed and proved an extended attack for multi-stage blockchain

August 2018 - August 2020

*New Delhi, India***ACADEMIC COMMUNITY WORK**

Tertiary reviewer for SIGGRAPH Asia 2025 and Pacific Graphics (PG) 2025

July 2025

Volunteered as a teaching assistant for the Poisson Reconstruction Project at Summer Geometry Initiative (SGI) 2024.

July 2024 - August 2024

Member of the Conference Coffee team at SIGGRAPH Research Career Development Committee.

2021-2022

TEACHING EXPERIENCE

University of Southern California

Teaching Assistant: CSCI 103 - Introduction to Programming

Fall 2025

Teaching Assistant: CSCI 580 - 3D Graphics and Rendering

Spring 2025

Teaching Assistant: CSCI 596 - Scientific Computing and Visualization

Fall 2024

Teaching Assistant: CSCI 104 - Data Structures and Object Oriented Design

Spring 2024

New York University

Grader: MATH 263.3 - Applied Partial Differential Equations

Spring 2023

Grader: MATH 252 - Numerical Analysis

Spring 2023

Grader: MATH 263 - Partial Differential Equations

Fall 2021